EVIDENCE AND FRAMING AUDITOR

The Ultimate Plain-English Explainer Guide for Peers & Friends

B.Tech CSE (AIML) Final Year Major Capstone Project · Lloyd Institute of Engineering & Technology
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1. The 60-Second Elevator Pitch: What Does This Project Do?

Imagine an AI that doesn't just give you a lazy *'Yes, this looks real'* or *'No, this is fake'*. Instead, imagine a **Digital Forensic News Auditor** that reads an article just like an expert investigative journalist would.

- ■ **It dissects the writing style:** Measures vocabulary richness, emotional hype, and whether the author is desperately trying to panic you.
- ■■ **It plots the political bias on a 2D map:** Shows whether the piece leans Left, Right, or Center, and calculates whether it is critiquing or praising specific policies.
- ■ **It catches sneaky logical fallacies:** Flags cheap tactics like personal insults (*Ad Hominem*), scary doomsday claims (*Appeal to Fear*), or panic triggers (*'share before deleted!'*).
- ■■ **It checks every single sentence against global authorities:** Cross-references medical claims with the World Health Organization (WHO), asteroid news with NASA, climate claims with the UN IPCC, and pulls live summaries from Wikipedia!
- ■ **It runs a Stacking Machine Learning Ensemble:** Fuses 5,028 linguistic features to deliver a calibrated **98.33% accuracy** score with full sentence-by-sentence proof.

> **■ In One Sentence:**
> *'Evidence and Framing Auditor is an explainable AI system that reads any news article or web URL, proves why it is authentic or deceptive, flags logical manipulation tricks, reveals hidden political bias, and verifies facts line-by-line against global institutional consensus.'*

2. Why Did We Build This? (The Misinformation Trap)

Most people think fake news is just bad grammar and typos. **That was true ten years ago.** Today, professional propagandists and generative AI tools write grammatically flawless articles that sound 100% convincing. Here is why the tools your friends usually use completely fail:

Existing Tool	How It Works	Why It Fails	How Our Project Fixes It
Keyword Blacklists	Searches for banned words like 'shocking' or 'secret'.	Smart writers easily swap synonyms ('unexpected', 'confidential').	We calculate 18+ mathematical stylometric metrics like Type-Token Ratio & Readability ease.

ChatGPT / LLMs	Generates an opinion or summarizes text.	Hallucinates fake citations, has high latency, and gives no verifiable math.	100% deterministic mathematical explainability (ABS, FSI, Bias_x) with real URLs.
Standard Media Bias Sites	Labels entire news outlets (e.g., 'CNN is Left, Fox is Right').	Misses individual articles! An article on Fox might praise a progressive bill.	Our **Target-Stance Engine** evaluates the specific article's praise vs criticism.
Open-Web Search Fact Checkers	Googles the headline and pulls random web results.	Pulls conflicting blogs and unverified social media opinions.	Directly checks hard data points from accredited global bodies (WHO, NASA, IPCC).

3. How It Works Behind The Scenes: The 5-Layer Secret Sauce

Whenever you paste an article or submit a live URL, our system runs it through a **5-Layer Pipeline** in under 3 seconds:

Layer 0: Live Ingestion: Paste raw text, drop a web link (parsed via BeautifulSoup4), or select live BBC World News RSS feeds.

Layer 1: Normalization & Preprocessing: Expands contractions (*can't -> cannot*), cleans unicode junk, splits sentences, and applies WordNet Lemmatization.

Layer 2A: Stylometrics & Readability: Computes Type-Token Ratio (vocabulary diversity), Simpson's Index, Flesch Reading Ease (FRE), FKGL grade level, and Punctuation Entropy.

Layer 2B: Target-Stance Ideological Bias: Scans political named entities and sentiment polarity to determine who is being criticized or praised, plotting an AllSides 2D coordinate.

Layer 2C: 8-Class Fallacy Detection: Scans for Ad Hominem smears, Appeals to Fear, False Dilemmas, and Conspiracy Framing, computing the Fallacy Severity Index (FSI).

Layer 2D: Institutional Knowledge Grounding: Audits every line against verified medical, planetary, climate, and cultural datasets (WHO, NASA, IPCC, UNESCO) + Wikipedia API.

Layer 3: Hybrid Stacking ML Ensemble: Concatenates 4,000 TF-IDF n-grams with 28 dense linguistic features (D = 5,028) into Logistic Regression + Random Forest + XGBoost.

Layer 4: Real-Time Telemetry Dashboard: Flask REST API on Port 5001 renders the final Verdict, AllSides 2D pin, claim table, and flagged high-risk deceptive sentences.

4. The Coolest Features You Can Show to Anyone

1. ■■ The Target-Stance Fix: Solving the Political Stance Inversion

The Problem: Old models look at words. If an article says: *'The radical left is completely destroying our economy'*, a naive AI sees the word 'left' and negative sentiment, and screams: *'This article is Left-Biased!'*

Our Fix: Our engine asks: *WHO is being attacked?* Here, the Left is the **target of criticism**! Criticizing Left policies means the author is coming from a **Right-leaning perspective**. Our algorithm correctly assigns a positive X coordinate (Right leaning) and explains it to the reader in plain English.

2. ■ The 8-Class Logical Fallacy Scanner

Instead of just saying an article is biased, it catches the exact psychological tricks being played on the reader:

- **Appeal to Fear:** *'Catastrophic apocalypse is coming, run for your lives!'*
- **Ad Hominem:** *'Don't listen to him, he is a corrupt puppet and a bought shill.'*
- **Phantom Authority:** *'Top anonymous whistleblowers secretly confirm the truth.'*
- **Manufactured Urgency:** *'Share this before it gets banned and deleted!'*

Each trick is highlighted with its exact sentence number and assigned a weighted **Fallacy Severity Index (FSI)**.

3. ■■ Claim-by-Claim Institutional Fact Grounding

Every single sentence is checked against an authoritative scientific knowledge base:

• If an article says *'baking soda cures all cancer overnight'* $ightarrow$ Tagged **[REFUTED]** with a direct citation to the **World Health Organization (WHO) & National Cancer Institute (NCI)** explaining why cancer involves 100+ genomic cellular types requiring multi-modal oncology.

• If an article says *'NASA is hiding an asteroid impact next week'* $ightarrow$ Tagged **[REFUTED]** citing **NASA CNEOS Sentry radar telemetry** confirming zero catastrophic impact risks for 100+ years.

4. ■ The AllSides 2D Spectrum Animated Pin

Just like the famous AllSides and Ad Fontes media charts, every article gets plotted on a 2D interactive canvas:

• **X-Axis (Ideological Lean):** From **-1.0 (Far Left)** to **+1.0 (Far Right)**.

• **Y-Axis (Factuality & Reliability):** From **0.0 (Low Factuality / High Spin)** to **1.0 (High Factuality / High Quality)**.

A glowing, animated pin drops right where the article lands!

5. Real Examples: What Happens When You Test It?

Here is what happens when you feed real-world headlines into the Evidence and Framing Auditor:

Headline / Article Snippet	Deception Verdict	Bias & 2D Position	Fallacies & Institutional Proof
'Secret household ingredient destroys all cancer tumors overnight! Doctors don\'t want you to know!'	**DECEPTIVE** Credibility: 15%	**High Distortion** (Bias: 0.0, Fact: 0.18)	• Fallacies: Phantom Authority + False Urgency • Grounding: **[REFUTED]** citing WHO Oncology Fact Sheet.
'The radical progressive agenda is a catastrophic disaster ruining freedom. Insiders say tyranny is here.'	**HIGH BIAS** Credibility: 42%	**Right / Conservative Stance** (Bias: +0.72, Fact: 0.35)	• Fallacies: Appeal to Fear + Ad Hominem • Stance: Adversarial Critique of Left-Wing Policies.
'NASA\'s James Webb Space Telescope discovers oldest known galaxy cluster emitting early cosmic light.'	**AUTHENTIC** Credibility: 98%	**Balanced Center** (Bias: 0.02, Fact: 0.96)	• Fallacies: None Detected (FSI: 0) • Grounding: **[SUPPORTED]** conforming to NASA astronomical telemetry.

6. How Can You Run and Try It Right Now?

You don't need fancy cloud servers or expensive GPUs. It runs locally on any standard computer in three simple steps:

1. **Open Terminal / PowerShell** and go to the project directory:
   `cd D:\Major_Project_FakeNewsDetection_Copy`
2. **Start the Application on Port 5001:**
   `python app.py 5001`
3. **Open Your Web Browser:**
   Visit **`http://127.0.0.1:5001`** and you'll see the full live dashboard!
   Click on any of the **Live Internet Presets** (powered by live BBC RSS feeds) or paste any article URL to see the system audit it live!

■ **The Capstone Engineering Team**

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